CSEC Mathematics

January 2014 – Paper 2

Solutions

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SECTION I

Answer ALL questions in this section.

All working must be clearly shown.

1. (a) Calculate the EXACT value of

$$\left(1\frac{3}{4}-\frac{1}{8}\right)+\left(\frac{5}{6}\div\frac{2}{3}\right)$$ [3]

$$\left(1\frac{3}{4}-\frac{1}{8}\right)+\left(\frac{5}{6}\div\frac{2}{3}\right)$$

$$=\left(\frac{7}{4}-\frac{1}{8}\right)+\left(\frac{5}{6}\div\frac{2}{3}\right)$$

$$=\left(\frac{14-1}{8}\right)+\left(\frac{5}{6}\times\frac{3}{2}\right)$$

$$=\left(\frac{13}{8}\right)+\left(\frac{5}{4}\right)$$

$$=\frac{13}{8}+\frac{5}{4}$$

$$=\frac{13+10}{8}$$

$$=\frac{23}{8}$$ (in exact form)

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(b) Using a calculator or otherwise, calculate

$$\sqrt{2.891} + \frac{1.2}{(1.31)^2}$$

giving your answer correct to 2 decimal places. [3]

Using a calculator,

$$\sqrt{2.891} + \frac{1.2}{(1.31)^2}$$

$$= 1.70 + \frac{1.2}{1.72}$$

$$= 1.70 + 0.70$$

$$= 2.40 \qquad \text{(correct to 2 decimal places)}$$

(c) A jeweller paid \$6 800 for 165 bracelets bought in China. The Customs Department in his country charged him \$1 360 in duty.

(i) Calculate the TOTAL cost of the 165 bracelets inclusive of duty. [1]

Total Cost = Cost of the bracelets + Duty fee

= 6800 + 1360

= \$8160

∴ The total cost of the 165 bracelets inclusive of duty is \$8160.

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(ii) The jeweller sold the 165 bracelets at a selling price of \$68.85 EACH.

(a) Calculate the TOTAL profit he made on the sale of the 165 bracelets.

[2]

Selling Price of 165 bracelets $= 165 \times \$68.85$

$= \$11360.25$

Since Selling Price (\$11360.25) is more than Cost Price (\$8160), a Profit was made.

Profit $=$ Selling Price $-$ Cost Price

$= \$11360.25 - \$8160$

$= \$3200.25$

(b) Calculate the profit as a percentage of the cost price, giving your answer to the nearest whole number. [2]

$$\text{Percentage Profit} = \frac{\text{Profit}}{\text{Cost price}} \times \frac{100}{1}$$

$$= \frac{3200.25}{8160} \times \frac{100}{1}$$

$$= 39.22$$

$$\approx 39\% \quad \text{(correct to the nearest whole number)}$$

Total: 11 marks

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2. (a) (i) Solve for x, where x is a real number

$$2(x-6)+3x \leq 8$$ [3]

$$2(x-6)+3x \leq 8$$

$$2x-12+3x \leq 8$$

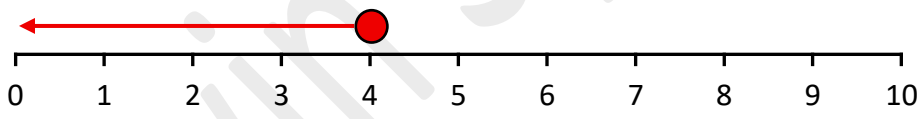
$$2x+3x \leq 8+12$$

$$5x \leq 20$$

$$x \leq \frac{20}{5}$$

$$x \leq 4$$

(ii) Using a number line, show your solution to Part (a)(i). [1]



(b) Factorize completely:

(i) $3x-6y+ax-2ay$ [2]

$$= 3x+ax-6y-2ay$$

$$= x(3+a)-2y(3+a)$$

$$= (x-2y)(3+a)$$

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(ii) $p^2 - 1$ [1]

$= (p)^2 - (1)^2$ [Difference of two squares]

$= (p-1)(p+1)$

(c) Expand and simplify

$(2k-3)(k-2)$ [2]

$(2k-3)(k-2)$

$= 2k^2 - 4k - 3k + 6$

$= 2k^2 - 7k + 6$

(d) The lines whose equations are

$3x + y = 2$ and $4x - 2y = 6$ intersect at (x, y)

Show that the point of intersection is $(1, -1)$. [3]

$3x + y = 2$ — Equation 1

$4x - 2y = 6$ — Equation 2

<u>Rearranging Equation 1 to get Equation 3:</u>

$3x + y = 2$

$y = 2 - 3x$ — Equation 3

Substiting Equation 3 into Equation 2:

$4x - 2(2 - 3x) = 6$

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$$4x - 4 + 6x = 6$$
$$10x = 6 + 4$$
$$10x = 10$$
$$x = \frac{10}{10}$$
$$x = 1$$

Substituting $x = 1$ into Equation 3 to find y:

$$y = 2 - 3(1)$$
$$y = 2 - 3$$
$$y = -1$$

$\therefore$ The point of intersection is $(1, -1)$.

Q.E.D.

Total: 12 marks

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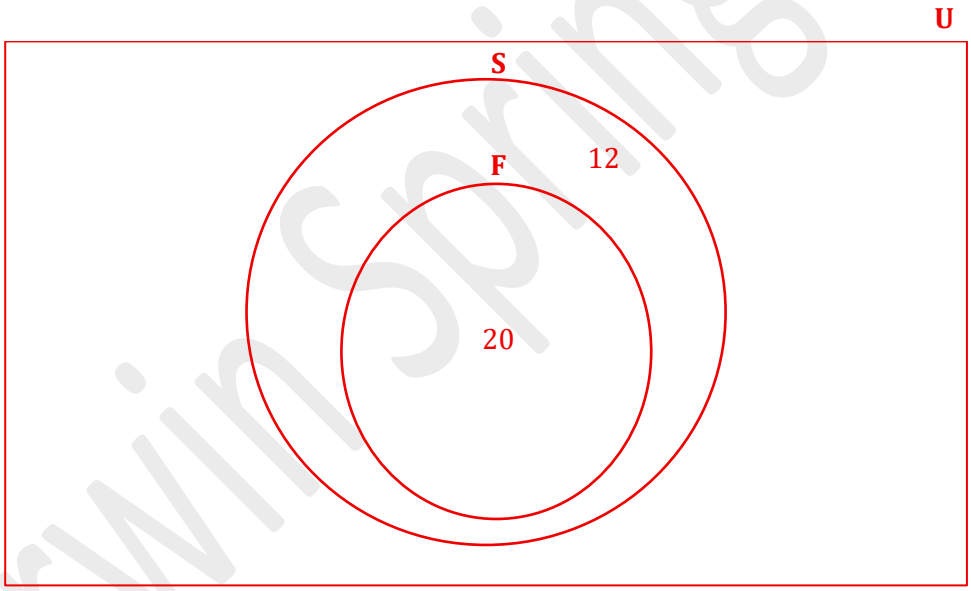
3. (a) In a class of 32 students, ALL of the students study Spanish (S) and 20 of the students study French (F).

(i) Represent this information on a Venn diagram. [2]

Let **U** = {Total students in the class}

Let **S** = {Students who study Spanish}

Let **F** = {Students who study French}



(ii) Calculate the number of students who study Spanish (S) but NOT French (F). [1]

Number of students who study Spanish (S) but NOT French (F)

$= 32 - 20$

$= 12$ students

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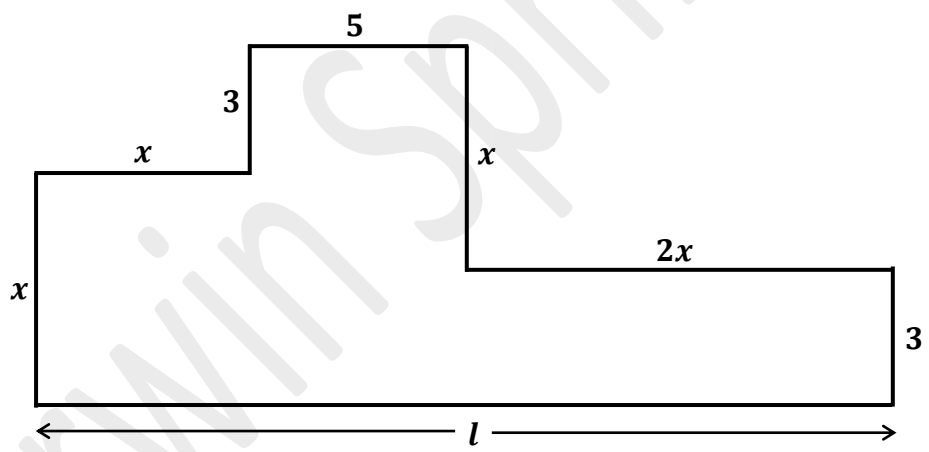
(iii) Write, using set notation, the relationship between F and S. [1]

F is a subset of S because all elements of F belong to S.

In set notation, this can be represented as:

$\boldsymbol{F \subset S}$

(b) The diagram below is an accurate plan of the floor of a room. All the corners are right angles. All measures shown in the diagram are recorded to the nearest metre.



(i) State, in terms of x, the length l of the floor. [1]

Length of the floor, $l = x + 5 + 2x$

$l = (3x + 5)\ m$

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(ii) The perimeter of the floor is 56 metres.

(a) Determine the value of x. [3]

$Length, l = (3x + 5)\ m$

$Width, w = (x + 3)\ m$

$$\begin{aligned} Perimeter &= 2(l + w) \\ &= 2((3x + 5) + (x + 3)) \\ &= 2(3x + 5 + x + 3) \\ &= 2(4x + 8) \\ &= 8x + 16 \end{aligned}$$

Since Perimeter was given as 56 m, we can say that

$$\begin{aligned} 8x + 16 &= 56 \\ 8x &= 56 - 16 \\ 8x &= 40 \\ x &= \frac{40}{8} \\ x &= 5 \end{aligned}$$

$\therefore x = 5\ m$

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(b) Calculate the area of the floor. [2]

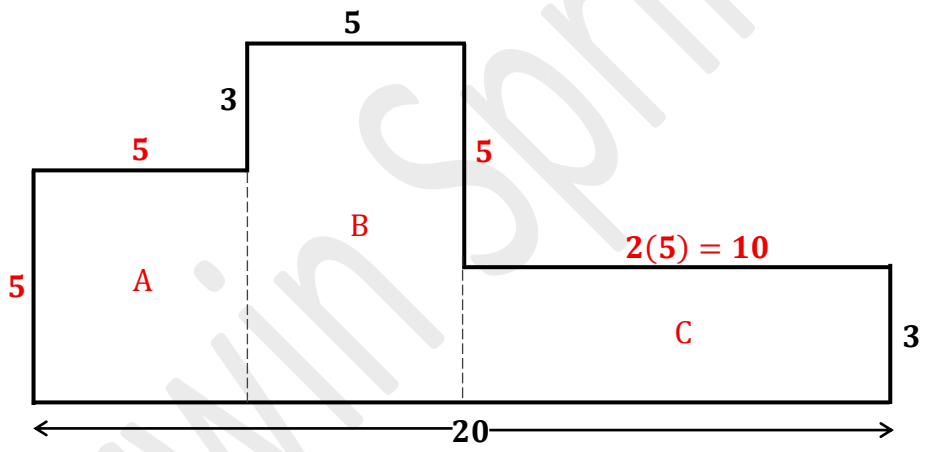
Since $x = 5\ m$,

$Length, l = (3x + 5)\ m$

$= (3(5) + 5)$

$= 15 + 5$

$= 20$



$Total\ Area \quad = Area\ of\ A + Area\ of\ B + Area\ of\ C$

$= (5 \times 5) + (5 \times 8) + (10 \times 3)$

$= 25 + 40 + 30$

$= 95\ m^2$

Total: 10 marks

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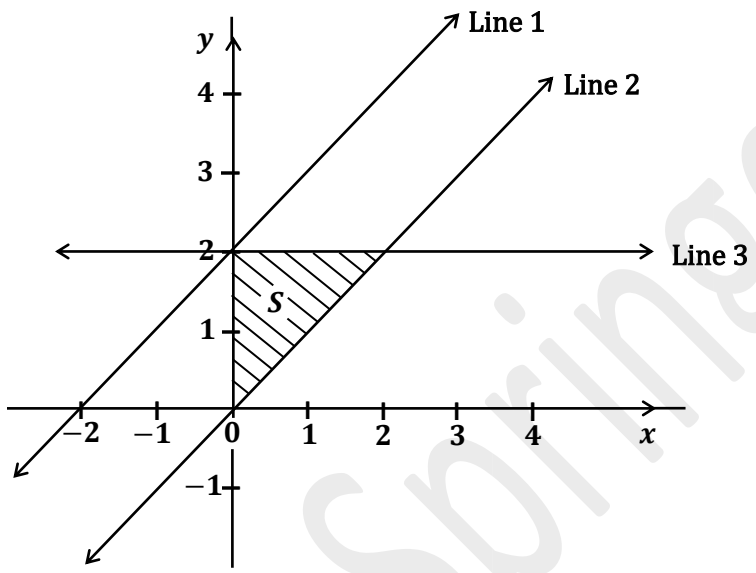
4. **An answer sheet is provided for this question.**

The diagram below shows the graphs of three straight lines:

$y = 2,$ $\quad y = x \quad$ and $\quad y = x + 2.$



On the answer sheet

(a) State the correct equation for EACH of the following:

(i) Line 1 [1]

Equation for Line 1:

$y = x + 2$ [Since Gradient = 1 and y-intercept = 2]

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(ii) Line 2 [1]

Equation for Line 2:

$y = x$ [Since Gradient $= 1$ and y-intercept $= 0$]

(iii) Line 3 [1]

Equation for Line 3:

$y = 2$ [Since Gradient $= 0$ and y-intercept $= 2$]

(b) Show that the gradient of Line 2 is 1. [1]

Two points that lie on Line 2 are: $(0, 0)$ and $(2, 2)$.

$$\text{Gradient} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{2 - 0}{2 - 0}$$

$$= \frac{2}{2}$$

$$= 1$$

Q.E.D

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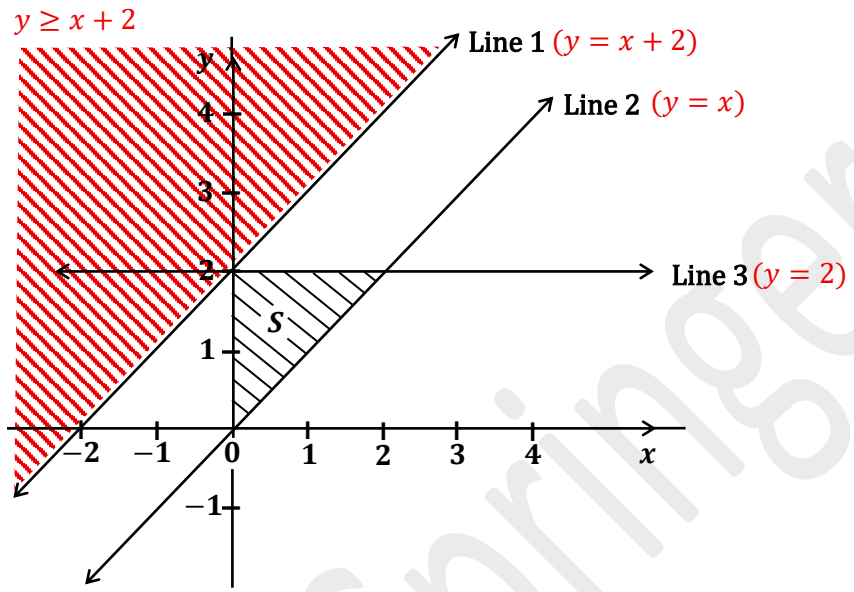
(c) Shade the region which is described as $y \geq x + 2$. [2]



(d) Write THREE inequalities that define the shaded region, S, shown in the diagram.

[3]

The three inequalities that define the shaded region, S are:

1) $y \leq 2$

2) $x \geq 0$

3) $y \geq x$

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(e) Write the equation of the straight line which is perpendicular to Line 1 and passes through the origin. [2]

Equation of Line 1: $y = x + 2$

The line that is perpendicular to Line 1 will have a gradient of -1. Since it passes through the origin $(0, 0)$, we can substitute $x_1 = 0$, $y_1 = 0$ and $m = -1$ into:

$y - y_1 = m(x - x_1)$

$y - 0 = -1(x - 0)$

$y = -1x$

$y = -x$

$\therefore$ The equation of the line that is perpendicular to Line 1 will be $y = -x$.

Total: 11 marks

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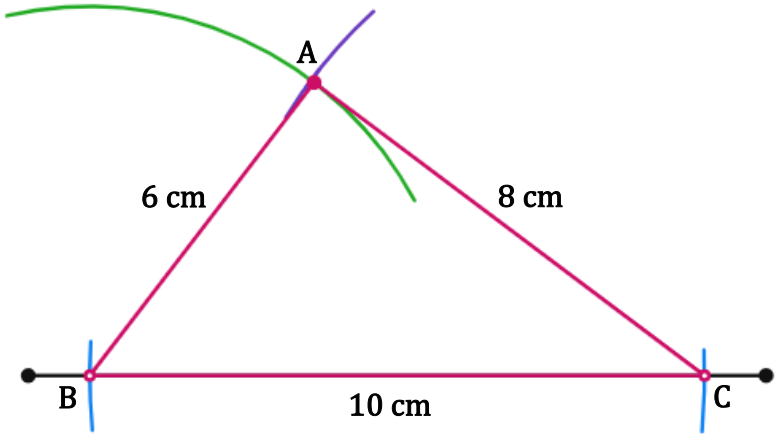
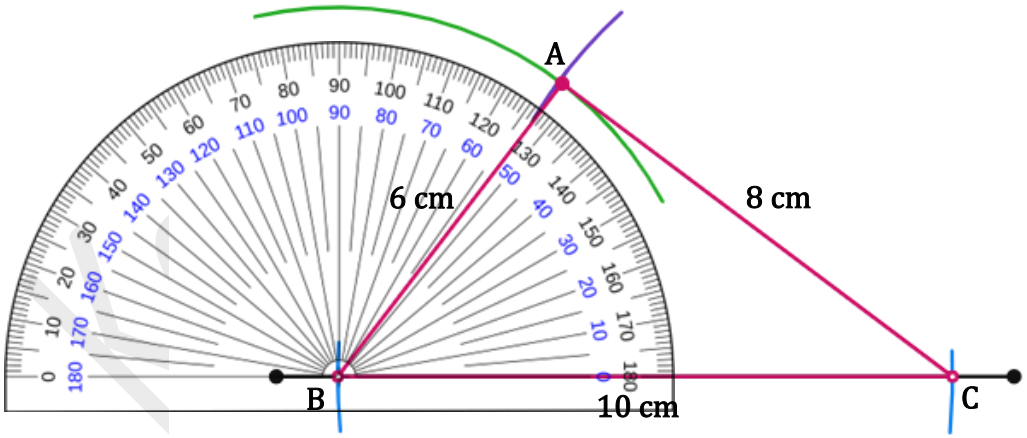
5. (a) (i) **Using a ruler**, **a pencil** and **a pair of compasses**, construct triangle ABC

with $BC = 10\ cm$, $AB = 6\ cm$ and $AC = 8\ cm$ [3]

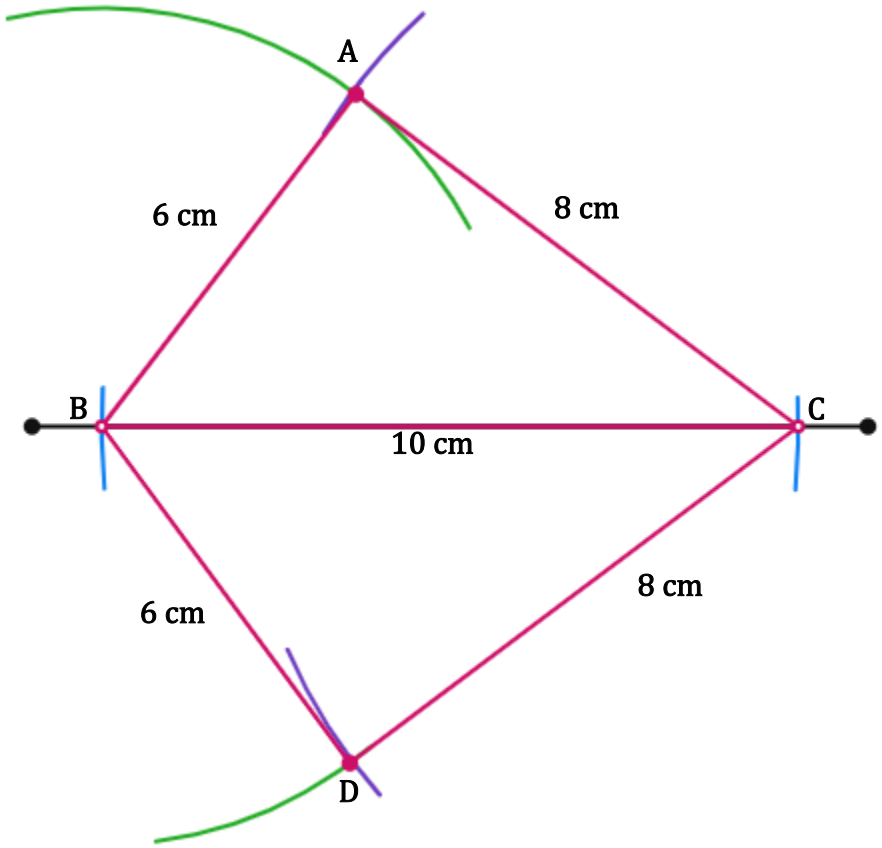


(ii) Measure and state the size of angle ABC. [1]



Angle $ABC = 53°$

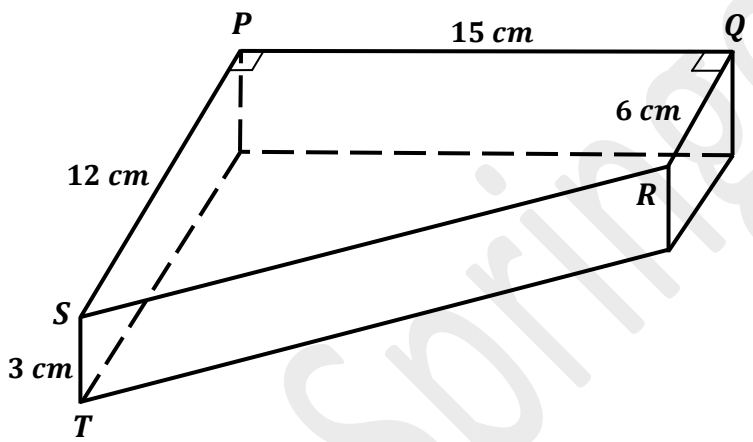
(iii) Complete the diagram to show a quadrilateral, $CABD$, such that $CD = CA$ and $BD = BA$. [2]



The quadrilateral, $CABD$ is a kite since the two pairs of adjacent sides of the quadrilateral are equal in length.

(b) The diagram below, **not drawn to scale**, shows a block of metal in the shape of a prism. The cross section of the block is a trapezium, $PQRS$, with parallel sides QR and PS, and PQ, the perpendicular distance between the sides.

$PQ = 15\ cm \qquad QR = 6\ cm \qquad PS = 12\ cm$ and $ST = 3\ cm$



(i) Calculate the area of the trapezium $PQRS$. [2]

In the trapezium above, the parallel sides are $PS = 12\ cm$ and $QR = 6\ cm$. The height $= 15\ cm$.

Substituting $a = 12\ cm$, $b = 6\ cm$ and $h = 15\ cm$ into:

$$Area\ of\ trapezium = \frac{1}{2}(a + b)h$$

$$= \frac{1}{2}(12 + 6) \times 15$$

$$= \frac{1}{2}(18) \times 15$$

$$= 9 \times 15$$

$$= 135\ cm^2$$

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(ii) Given that $ST = 3$ cm, calculate the volume of the block of metal. [1]

$$Volume = Area\ of\ Trapezium \times 3$$

$$= 135\ cm^2 \times 3\ cm$$

$$= 405\ cm^3$$

(iii) The block of metal has a mass of 1.5 kg. Calculate, in grams, the mass of ONE cubic centimetre of metal. [3]

Mass of the block $= 1.5$ kg

$= 1.5 \times 1000$

$= 1500$ g

Mass of 1 $cm^3 = \frac{1500}{405}\ g$

$= 3.7\ g$ [to 2 significant figures]

Total: 12 marks

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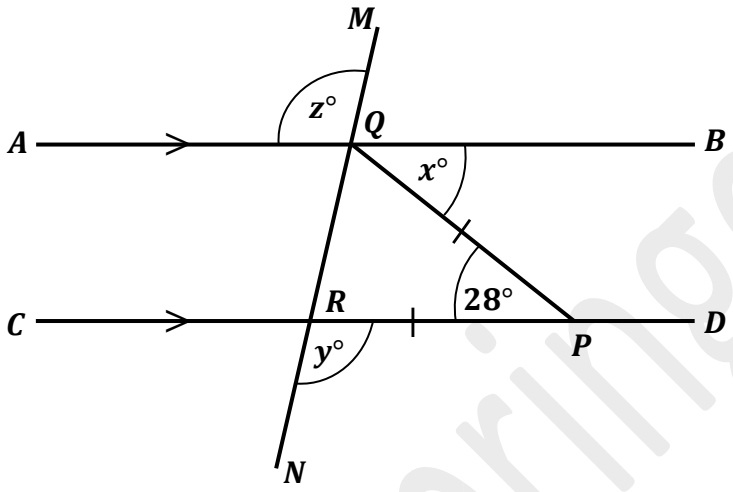
6. (a) In the figure below, **not drawn to scale**, the lines AQB and $CRPD$ are parallel and $MQRN$ is a transversal. $PQ = PR$ and angle $QPR = 28°$.



Calculate, giving reasons for your answer, the value of

(i) x [2]

$x = 28°$

Since the alternate interior angles are equal.

$B\hat{Q}P = Q\hat{P}R = 28°$.

(ii) y [2]

Triangle QPR is an isosceles triangle where the base angles are equal.

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$$P\hat{Q}R = Q\hat{R}P = \frac{180° - 28°}{2}$$

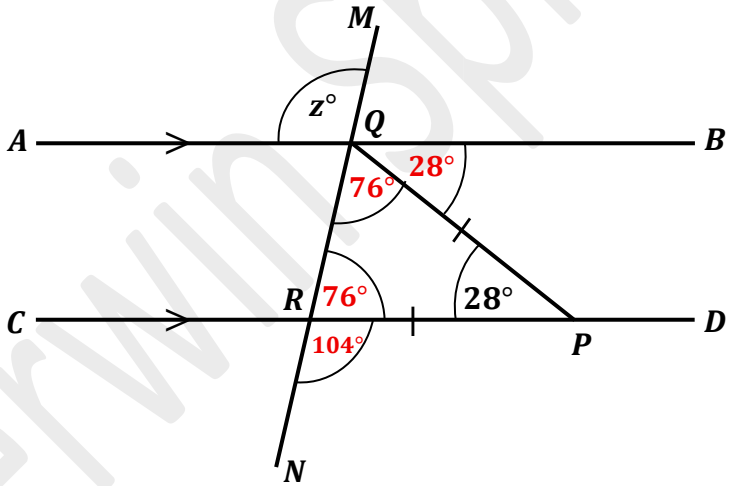
$$= \frac{152}{2}$$

$$= 76°$$

Since the angles on a straight line add up to 180°,

$y = 180° - 76°$

$= 104°$



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(iii) z [2]

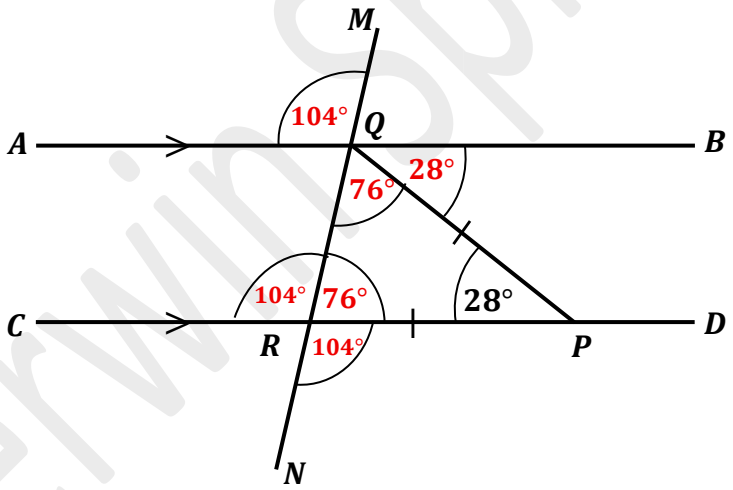
Since angles on a straight line add up to 180°,

$Q\hat{R}C = 180° - 76°$

$= 104°$

$z = Q\hat{R}C$ [since corresponding angles are equal]

$= 104°$



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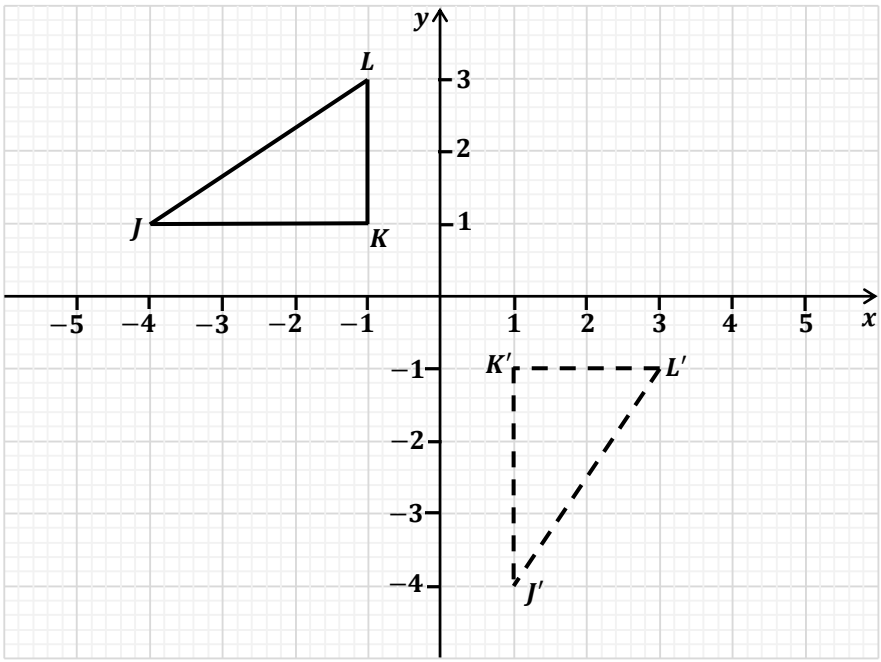
(b) The diagram below shows triangle JKL and its image $J'K'L'$ after a transformation.



(i) State the coordinates of the vertex, J, of triangle JKL. [1]

Reading off the graph,

$J = (-4, 1)$

(ii) State the length of the side $K'L'$ of triangle $J'K'L'$. [1]

Since 1 block = 1 cm,

$K'L' = 2\ blocks$

$= 2\ cm$

(iii) Describe FULLY a **single** transformation that maps triangle JKL onto triangle $J'K'L'$. [2]

The single transformation that maps triangle JKL onto triangle $J'K'L'$ is a relfection in the line $y = x$.

(iv) State the coordinates of triangle $J"K"L"$, the image of triangle JKL, under the translation by the vector $\begin{pmatrix} 5 \\ -3 \end{pmatrix}$. [2]

$J = \begin{pmatrix} -4 \\ 1 \end{pmatrix}$ $K = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ $L = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$

<u>The coordinates of triangle $J"K"L"$:</u>

$J'' = \begin{pmatrix} -4 \\ 1 \end{pmatrix} + \begin{pmatrix} 5 \\ -3 \end{pmatrix}$ $K'' = \begin{pmatrix} -1 \\ 1 \end{pmatrix} + \begin{pmatrix} 5 \\ -3 \end{pmatrix}$ $L'' = \begin{pmatrix} -1 \\ 3 \end{pmatrix} + \begin{pmatrix} 5 \\ -3 \end{pmatrix}$

$J'' = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$ $K'' = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$ $L'' = \begin{pmatrix} 4 \\ 0 \end{pmatrix}$

$\therefore$ The coordinates of triangle $J"K"L"$ are as follows:

$J'' = (1, -2)$

$K'' = (4, -2)$

$L'' = (4, 0)$

Total: 12 marks

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7. The heights of a sample of seedlings were measured to the nearest centimetre and then arranged in class intervals as shown in the table below.

Height in *cm*	Midpoint	Frequency
3 – 7	5	0
8 – 12	10	3
13 – 17	15	12
18 – 22		16
23 – 27		22
28 – 32		18
33 – 37		14

(a) How many seedlings were in the sample? [1]

Total number of seedlings $= 0 + 3 + 12 + 16 + 22 + 18 + 14$

$= 85$

(b) For the class interval written as "8 – 12" in the table above, write down

(i) the lower class **limit** [1]

For the class interval 8 – 12, the lower class limit is 8.

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(ii) the upper class **boundary** [1]

For the class interval $8 - 12$, the upper class boundary is 12.5.

(iii) the class **width** [1]

For the class interval $8 - 12$,

The lower class boundary $= 7.5$

The upper class boundary $= 12.5$

Class width $=$ Upper class boundary $-$ Lower class boundary

$= 12.5 - 7.5$

$= 5$

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(c) Copy and complete the table by inserting

(i) the midpoints of EACH class interval [2]

(ii) the missing values for the class interval **after** the interval "33 – 37". [1]

Height in *cm*	**Midpoint**	**Frequency**
3 – 7	5	0
8 – 12	10	3
13 – 17	15	12
18 – 22	$\frac{17.5 + 22.5}{2} = 20$	16
23 – 27	$\frac{22.5 + 27.5}{2} = 25$	22
28 – 32	$\frac{27.5 + 32.5}{2} = 30$	18
33 – 37	$\frac{32.5 + 37.5}{2} = 35$	14
38 – 42	$\frac{37.5 + 42.5}{2} = 40$	0

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(d) Using a scale of **2 *cm* to represent 5 *cm* on the horizontal axis** and **2 *cm* to represent 5 seedlings on the vertical axis**, draw a frequency polygon to represent the data as shown in your table at (c). [5]

Midpoint	Frequency	Points to be plotted
5	0	(5, 0)
10	3	(10, 3)
15	12	(15, 12)
20	16	(20, 16)
25	22	(25, 22)
30	18	(30, 18)
35	14	(35, 14)
40	0	(40, 0)

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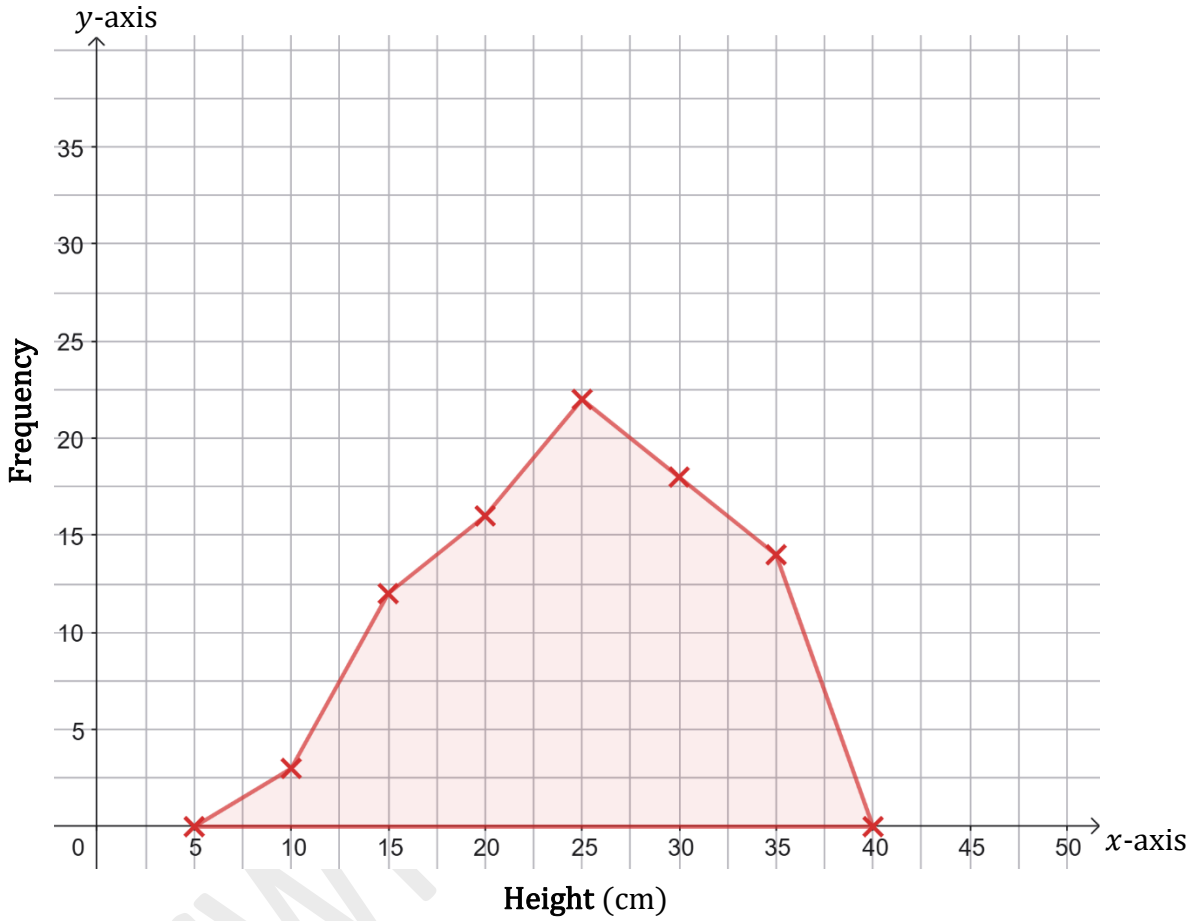
Title: Frequency Polygon Showing The Heights Of A Sample Of Seedlings



Scale:

x-axis: 2 cm represents 5 cm

y-axis: 2 cm represents 5 seedlings

Total: 12 marks

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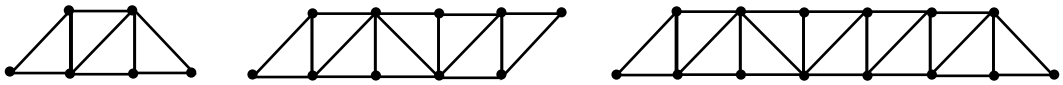
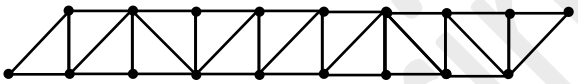
8.

Figure 1 Figure 2 Figure 3



(a) **On the answer sheet provided**, draw Figure 4, the FOURTH figure in the sequence. [2]

Figure 4



(b) Study the patterns in the table below. Complete the table by inserting the missing values in the rows numbered (i), (ii), (iii) and (iv).

	No. of Trapezia (n)	No. of Triangles	No. of Dots	
	1	4	6	
	2	8	10	
	3	12	14	
(i)	4	16	18	[2]
(ii)	10	40	42	[2]
(iii)	16	64	66	[2]
(iv)	n	$4n$	4n+2	[2]

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(iv) Looking at Figures 1, 2 and 3, we can deduce that

Number of Triangles $= 4n$

Number of Dots $= 4n + 2$

Explanation:

Number of Triangles:

No. of Trapezia (n)	No. of Triangles	No. of Dots
1	4	6
2	8	10
3	12	14

The number of triangles increase by 4.

$\therefore$ In terms of n, we can deduce that $Number\ of\ Triangles = 4n + k$

[where k is a constant]

Substituting $n = 1$ and $Number\ of\ Triangles = 4$ into:

$Number\ of\ Triangles = 4n + k$

$4 = 4(1) + k$

$4 = 4 + k$

$4 - 4 = k$

$0 = k$

$k = 0$

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We can also substitute $n = 2$ and Number of Triangles $= 8$ OR $n = 3$ and Number of Squares $= 12$ to confirm the value of k.

$\therefore$ Number of Triangles $= 4n$

Number of Dots:

No. of Trapezia (n)	No. of Triangles	No. of Dots
1	4	6
2	8	10
3	12	14

The number of dots increase by 4.

$\therefore$ In terms of n, we can deduce that $Number\ of\ Dots = 4n + k$

[where k is a constant]

Substituting $n = 1$ and $Number\ of\ Dots = 6$ into:

$Number\ of\ Dots = 4n + k$

$6 = 4(1) + k$

$6 = 4 + k$

$6 - 4 = k$

$2 = k$

$k = 2$

We can also substitute $n = 2$ and Number of Dots $= 10$ OR $n = 3$ and Number of Squares $= 14$ to confirm the value of k.

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$\therefore$ Number of Dots $= 4n + 2$

(i) Number of Triangles $= 4n$

$= 4(4)$

$= 16$

Number of Dots $= 4n + 2$

$= 4(4) + 2$

$= 16 + 2$

$= 18$

(ii) When $n = 10$,

Number of Triangles $= 4(10)$

$= 40$

Number of Dots $= 4n + 2$

$= 4(10) + 2$

$= 40 + 2$

$= 42$

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(iii) Number of triangles $= 64$,

Number of Triangles $= 4n$,

$64 = 4n$

$\frac{64}{4} = n$

$n = 16$

When $n = 16$,

Number of Dots $= 4(16) + 2$

$= 66$

Total: 10 marks

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SECTION II

Answer TWO questions in this section.

ALGEBRA AND RELATIONS, FUNCTIONS AND GRAPHS

9. (a) The functions $h(x)$ and $g(x)$ are defined as

$$h(x) = \frac{10}{x} - 3 \qquad\qquad g(x) = 3x - 2.$$

(i) Evaluate:

(a) $g(4)$ [1]

$$g(x) = 3x - 2$$

$$g(4) = 3(4) - 2$$

$$= 12 - 2$$

$$= 10$$

(b) $hg(4)$ [2]

$$hg(x) = h[3x - 2]$$

$$= \frac{10}{3x - 2} - 3$$

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$$hg(4) = \frac{10}{3(4) - 2} - 3$$

$$= \frac{10}{12 - 2} - 3$$

$$= \frac{10}{10} - 3$$

$$= 1 - 3$$

$$= -2$$

(ii) Write expressions in x for:

(a) $h^{-1}(x)$

$$h(x) = \frac{10}{x} - 3$$

Let $y = h(x)$

$$y = \frac{10}{x} - 3$$

Make x the subject of the formula:

$$y = \frac{10}{x} - 3$$

$$y + 3 = \frac{10}{x}$$

$$x = \frac{10}{y + 3}$$

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Interchange x and y:

$$y = \frac{10}{x+3}$$

$$\therefore h^{-1}(x) = \frac{10}{x+3}$$

(b) $gg(x)$ [4]

$$g(x) = 3x - 2$$

$$\begin{aligned} gg(x) &= g[g(x)] \\ &= g[3x-2] \\ &= [3(3x-2)-2] \\ &= [(9x-6)-2] \\ &= [9x-6-2] \\ &= 9x-8 \end{aligned}$$

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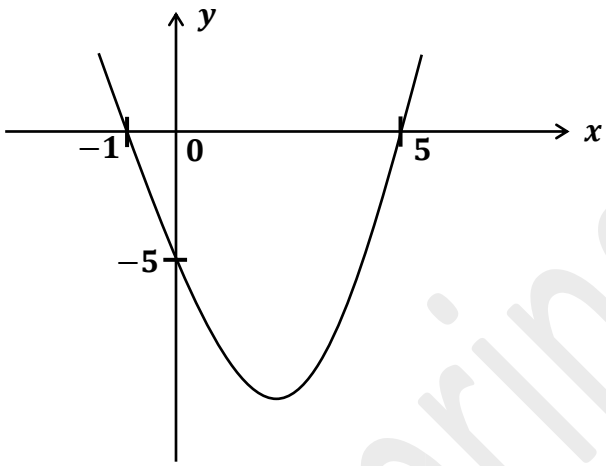
(b) The diagram below shows a sketch of the quadratic function

$$y = x^2 + bx + c, \qquad -2 \leq x \leq 6$$

where b and c are constants.



(i) State the roots of the equation $x^2 + bx + c = 0$. [2]

Reading off the diagram, the roots occur when:

$x = -1$ and $x = 5$.

(ii) For the function $y = x^2 + bx + c$

(a) determine the value of c [2]

Reading off the graph, the value of c for the function $y = x^2 + bx + c$ is -5.

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(b) show that $b = -4$ [2]

Since roots occur when $x = -1$ and $x = 5$,

$x^2 + bx + c = (x + 1)(x - 5)$

$= x^2 - 4x - 5$

$\therefore$ Equating the coefficients, we get $b = -4$.

Q.E.D.

(iii) State the coordinates of the MINIMUM point on the graph of the function $y = x^2 + bx + c$. [2]

$y = x^2 - 4x - 5$

The minimum point occurs when x is halfway between the roots. In other words,

$x = \frac{-1 + 5}{2}$

$x = \frac{4}{2}$

$x = 2$

When $x = 2$,

$y = (2)^2 - 4(2) - 5$

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$y = 4 - 8 - 5$

$y = -9$

$\therefore$ The coordinates of the minimum point is $(2, -9)$.

Total: 15 marks

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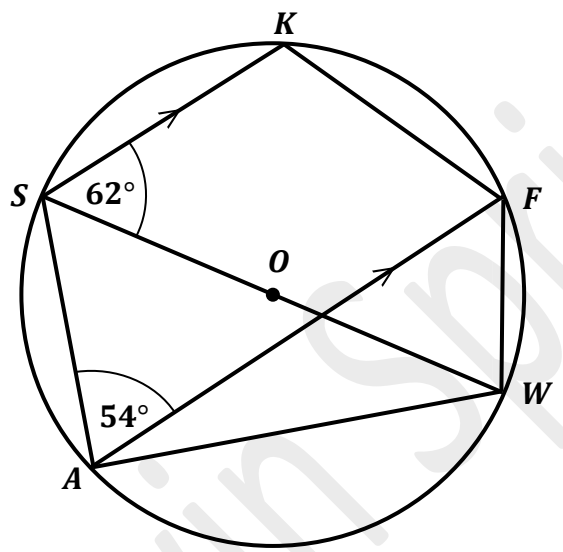
10. (a) In the diagram below, **not drawn to scale**, O is the centre of the circle. The lines SK and AF are parallel.

$\angle KSW = 62°$ $\angle SAF = 54°$



Calculate, **giving reasons for your answer**, the measure of:

(i) $\angle FAW$ [2]

$S\hat{A}W = 90°$ [since the angle in a semi-circle is 90°].

$F\hat{A}W = 90° - 54°$

$= 36°$

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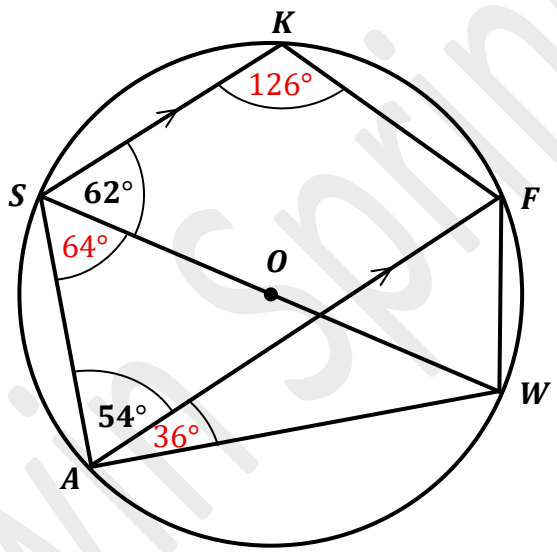
(ii) $\angle SKF$ [2]

$S\hat{K}F = 180° - 54°$

$= 126°$ [Since the opposite angles in a cyclic quadrilateral add up to 180°]

(iii) $\angle ASW$ [2]



$K\hat{S}A$ and $S\hat{A}F$ are co-interior opposite angles and are therefore supplementary (add up to 180°)

$A\hat{S}W = 180° - (62° + 54°)$

$= 64°$

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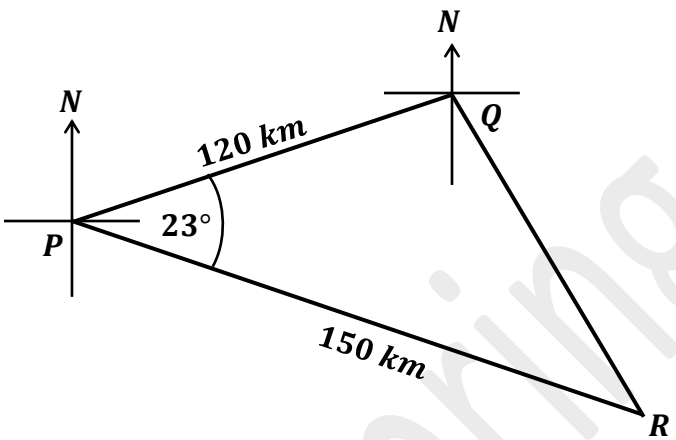
(b) The diagram below, **not drawn to scale**, shows the position of three points, P, Q and R on a horizontal plane.

$PQ = 120\ km$ $\quad$ $PR = 150\ km$ $\quad$ $\angle QPR = 23°$



(i) Calculate, correct to one decimal place

(a) the distance QR [3]

Using the cosine rule,

$QR^2 = PQ^2 + PR^2 - 2(PQ)(PR)\cos P$

$= (120)^2 + (150)^2 - 2(120)(150)\cos(23°)$

$= 14400 + 22500 - 36000\cos(23°)$

$= 3761.83$

$QR = \sqrt{3761.83}$

$= 61.3$ km [correct to 1 decimal place]

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(b) the area of triangle PQR [2]

$$Area\ of\ PWR = \frac{1}{2}\,(PQ)(PR)(\sin P)$$

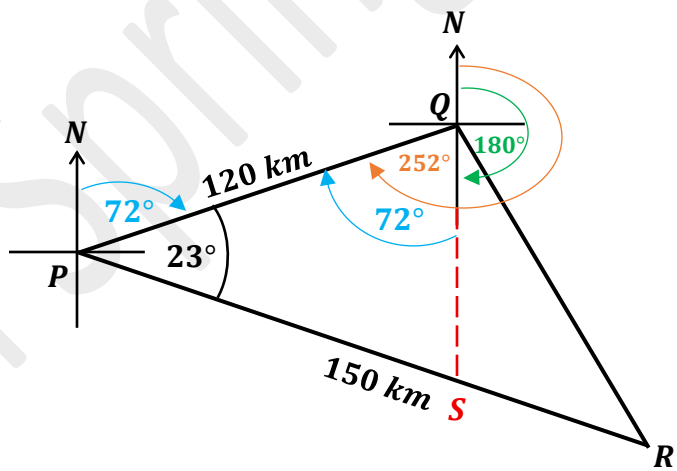
$$= \frac{1}{2}\,(120)(150)(\sin 23°)$$

$$= 3516.6\ km^2 \qquad \text{[correct to 1 decimal place]}$$

(ii) The bearing of P from Q is 252°. Calculate the bearing of R from P. [4]

$P\hat{Q}S = 252° - 180°$

$= 72°$



$N\hat{P}R = 72° + 23°$

$= 95°$

∴ The bearing of R from P is 095°.

Total: 15 marks

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11. (a) The matrix, T, is such that

$$T = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}$$

(i) Determine, T^{-1}, the inverse of T. [3]

$T = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}$ which is of the form

$T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

Where $a = 2, b = -1, c = 1, d = 3.$

$$T^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$= \frac{1}{(2)(3) - (-1)(1)}\begin{pmatrix} 3 & -(-1) \\ -(1) & 2 \end{pmatrix}$$

$$= \frac{1}{6 + 1}\begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix}$$

$$= \frac{1}{7}\begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{3}{7} & \frac{1}{7} \\ -\frac{1}{7} & \frac{2}{7} \end{pmatrix}$$

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(ii) The matrix T maps the point (a, b) onto the point $(4, 9)$. Determine the values of a and b. [4]

If (a, b) is mapped onto (4,9) under the translation T, then (4,9) would be mapped onto (a, b) under the translation T^{-1}, therefore we multiply the point (4,9) by the inverse of T to yield the values of a and b respectfully.

$$\begin{pmatrix} \frac{3}{7} & \frac{1}{7} \\ -\frac{1}{7} & \frac{2}{7} \end{pmatrix}\begin{pmatrix} 4 \\ 9 \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix}$$

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \frac{3}{7} & \frac{1}{7} \\ -\frac{1}{7} & \frac{2}{7} \end{pmatrix}\begin{pmatrix} 4 \\ 9 \end{pmatrix}$$

$$= \begin{pmatrix} \left(\frac{3}{7} \times 4\right) + \left(\frac{1}{7} \times 9\right) \\ \left(-\frac{1}{7} \times 4\right) + \left(\frac{2}{7} \times 9\right) \end{pmatrix}$$

$$= \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

$\therefore a = 3, b = 2$

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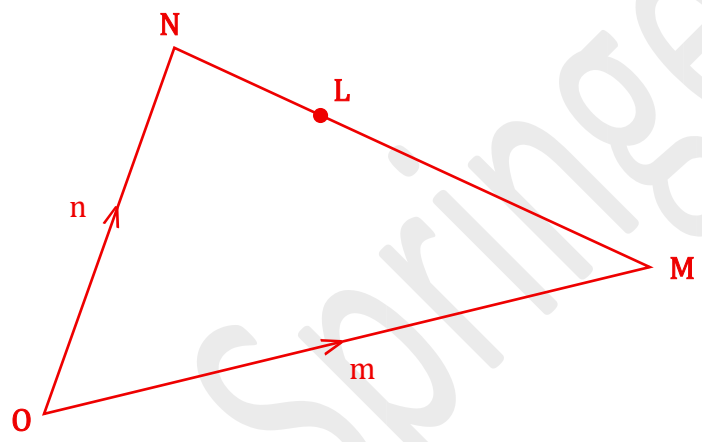
(b) $\overrightarrow{OM}$ and $\overrightarrow{ON}$ are position vectors with respect to the origin, O, such that

$\overrightarrow{OM} = \boldsymbol{m}$ and $\overrightarrow{ON} = \boldsymbol{n}$

L is a point on MN such that $ML : LN = 2 : 1$.

(i) Draw a sketch of the triangle OMN and label the points O, M, N and L. [2]



(ii) Write in terms of $\boldsymbol{m}$ and $\boldsymbol{n}$ an expression for:

(a) $\overrightarrow{MN}$

$$\overrightarrow{MN} = \overrightarrow{MO} + \overrightarrow{ON}$$

$$= -m + n$$

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(b) $\overrightarrow{ML}$ [3]

$$\overrightarrow{ML} = \frac{2}{3}\overrightarrow{MN}$$

$$= \frac{2}{3}(-m + n)$$

(iii) If $\boldsymbol{m} = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$ and $\boldsymbol{n} = \begin{pmatrix} 9 \\ 0 \end{pmatrix}$,

determine the position vector of L. [3]

$$\overrightarrow{OL} = \overrightarrow{OM} + \overrightarrow{ML}$$

$$= m + \frac{2}{3}(-m + n)$$

$$= m - \frac{2}{3}m + \frac{2}{3}n$$

$$= \frac{1}{3}m + \frac{2}{3}n$$

$$= \frac{1}{3}\begin{pmatrix} 3 \\ 6 \end{pmatrix} + \frac{2}{3}\begin{pmatrix} 9 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 6 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 7 \\ 2 \end{pmatrix}$$

Total: 15 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

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